Fachhochschule Nordwestschweiz
Hochschule für Technik

Intro PyTorch

Deep Learning

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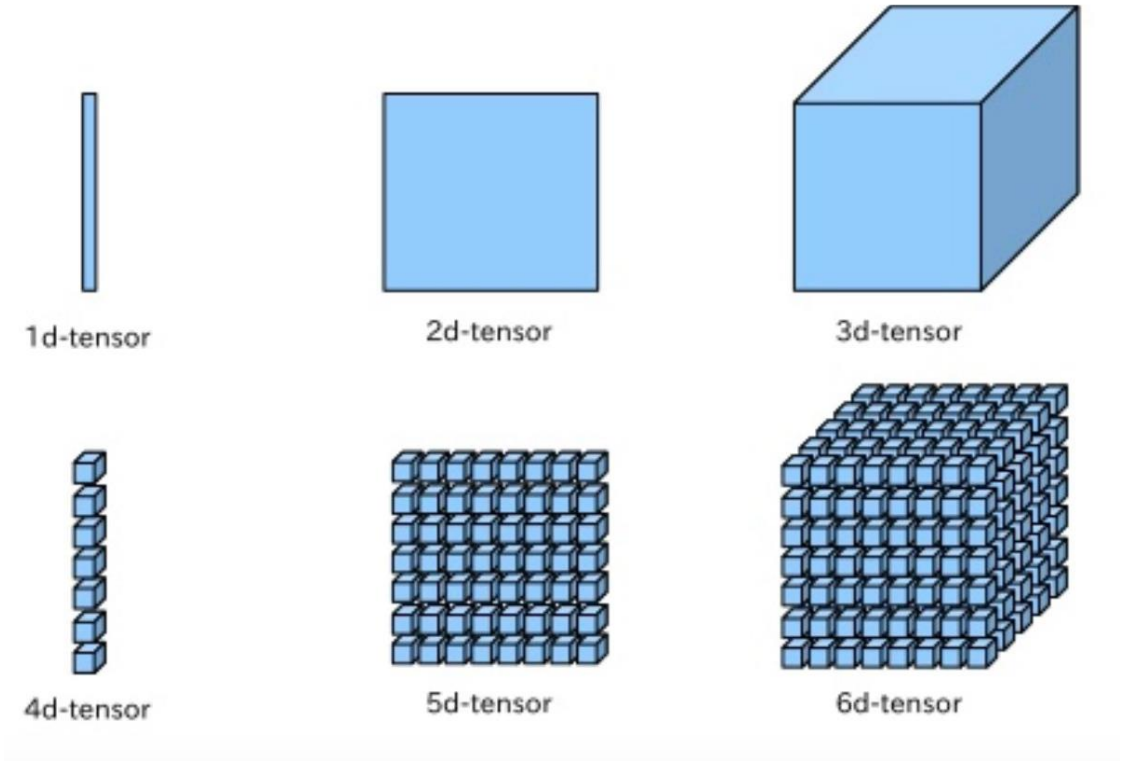
Martin Melchior

First Steps in PyTorch



Setup PyTorch and Jupyter

test in jupyter notebook

1. Create and activate new environment:

```
conda create --name del_hs25 python=3.9
conda activate del_hs25
```

3. Install jupyter lab, matplotlib, pandas:

```
conda install jupyterlab matplotlib pandas
```

2. Install PyTorch

now 3.10

NOTE: Latest PyTorch requires Python 3.9 or later.

now 2.10.0

PyTorch Build	Stable (2.8.0)			Preview (Nightly)	
Your OS	Linux	Mac		Windows	
Package	Pip	LibTorch		Source	
Language	Python			C++ / Java	
Compute Platform	~~CUDA 12.6~~	~~CUDA 12.8~~	~~CUDA 12.9~~	~~ROCm 6.4~~	Default
Run this Command:	`pip3 install torch torchvision`				

Possible issue with several OpenMP versions linked.

```
[1]: import torch

[2]: print (torch.__version__)
     2.8.0

[3]: print(torch.cuda.is_available())
     print(torch.backends.mps.is_available())
     False
     True
```

linux, windows

mac

PyTorch Tensors

- **Tensor** is a generalized matrix, a finite table of numerical values indexed along several discrete dimensions.
  - 0d tensor: a scalar.
  - 1d tensor: a vector (e.g. a sound sample).
  - 2d tensor: a matrix (e.g. a grayscale image).
  - 3d tensor: a vector of identically sized matrices (e.g. a RGB image).
  - 4d tensor: a matrix of identically sized matrices, or a sequence of 3d tensors (e.g. a sequence of RGB images, a video, dataset of images)

- Constrained structure of a tensor allows to be processed by using CPUs and GPUs, at [near] peak performance.
  - E.g. standard linear algebra operations, but also convolution, pooling, etc.

PyTorch Tensors

- Efficient tensor operations on CPU/GPU:
  - Very similar to numpy operations, incl broadcasting
  - standard linear algebra
  - deep-learning specific operations (conv, pooling)
- Automatic on-the-fly differentiation (autograd)
- Properties:
  - device: Tensors live on a device
  - dtype: by default `float32`
- Data I/O

Pytorch Cheat Sheet

Source: Francois Fleuret

Creation

- torch.empty(*size, ...)
- torch.zeros(*size, ...)
- torch.full(size, value, ...)
- torch.tensor(sequence, ...)
- torch.eye(n, ...)
- torch.from_numpy(ndarray)

Indexing, Slicing, Joining, Mutating

- torch.Tensor.view(*size)
- torch.cat(inputs, dimension=0)
- torch.chunk(tensor, nb_chunks, dim=0)[source]
- torch.split(tensor, split_size, dim=0)[source]
- torch.index_select(input, dim, index, out=None)
- torch.t(input, out=None)
- torch.transpose(input, dim0, dim1, out=None)

Filling

- Tensor.fill_(value)
- torch.bernoulli_(proba)
- torch.normal_([mu, [std]])

Pointwise math

- torch.abs(input, out=None)
- torch.add()
- torch.cos(input, out=None)
- torch.sigmoid(input, out=None)

Math reduction

- torch.dist(input, other, p=2, out=None)
- torch.mean()
- torch.norm()
- torch.std()
- torch.sum()

BLAS and LAPACK Operations

- torch.eig(a, eigenvectors=False, out=None)
- torch.lstsq(B, A, out=None)
- torch.inverse(input, out=None)
- torch.mm(mat1, mat2, out=None)
- torch.mv(mat, vec, out=None)

```
[1]: import torch
```

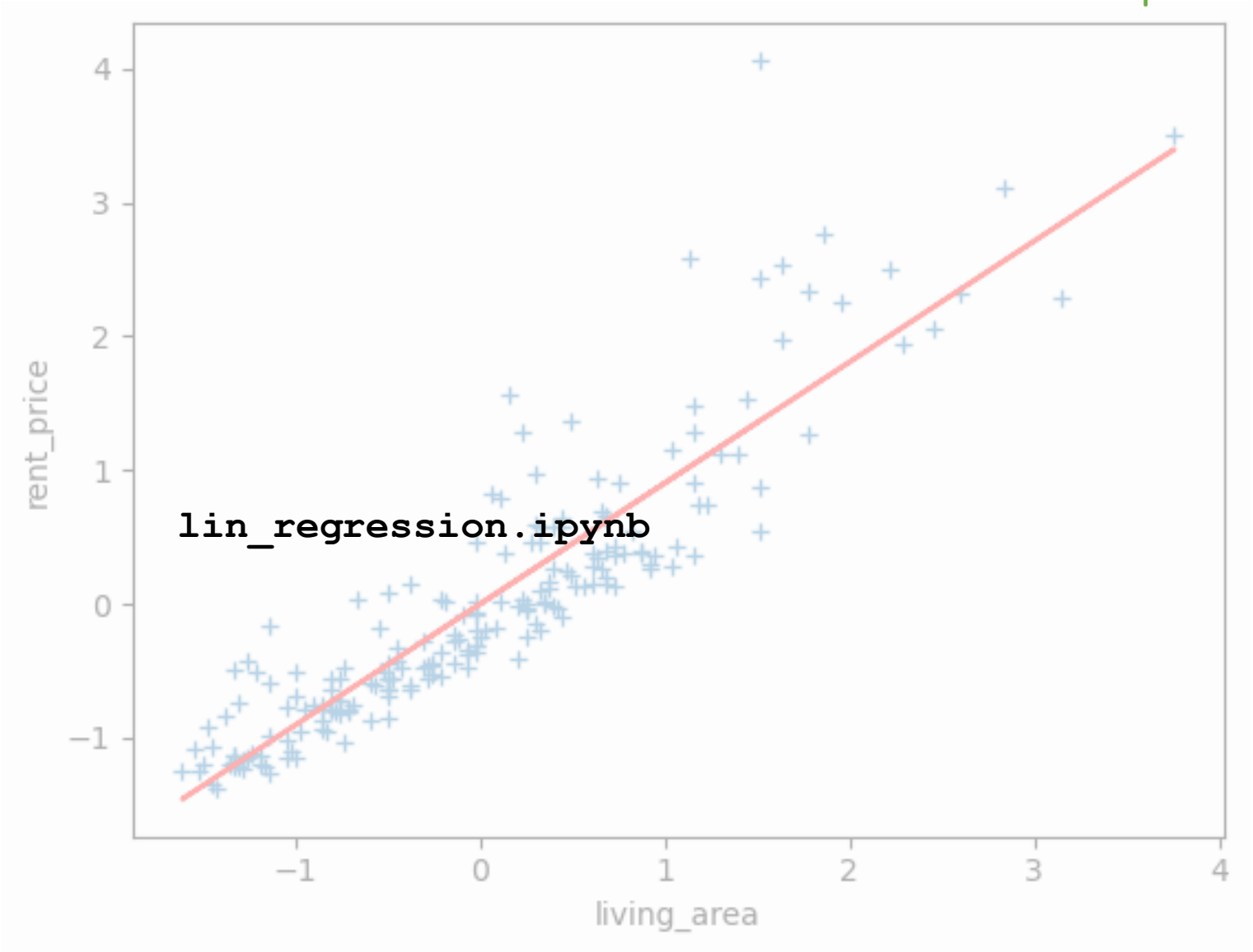
Initialization

```
[2]: # from lists
     x = torch.Tensor([5, 3])
     x, x.shape, x.dtype, x.device

[2]: (tensor([5., 3.]), torch.Size([2]), torch.float32, device(type='cpu'))

[3]: # with zeros or ones
     x = torch.zeros([3, 2])
     x, x.shape

[3]: (tensor([[0., 0.],
              [0., 0.],
              [0., 0.]]),
      torch.Size([3, 2]))

[4]: y = torch.ones([2, 3])
     y, y.shape

[4]: (tensor([[1., 1., 1.],
              [1., 1., 1.]]),
```

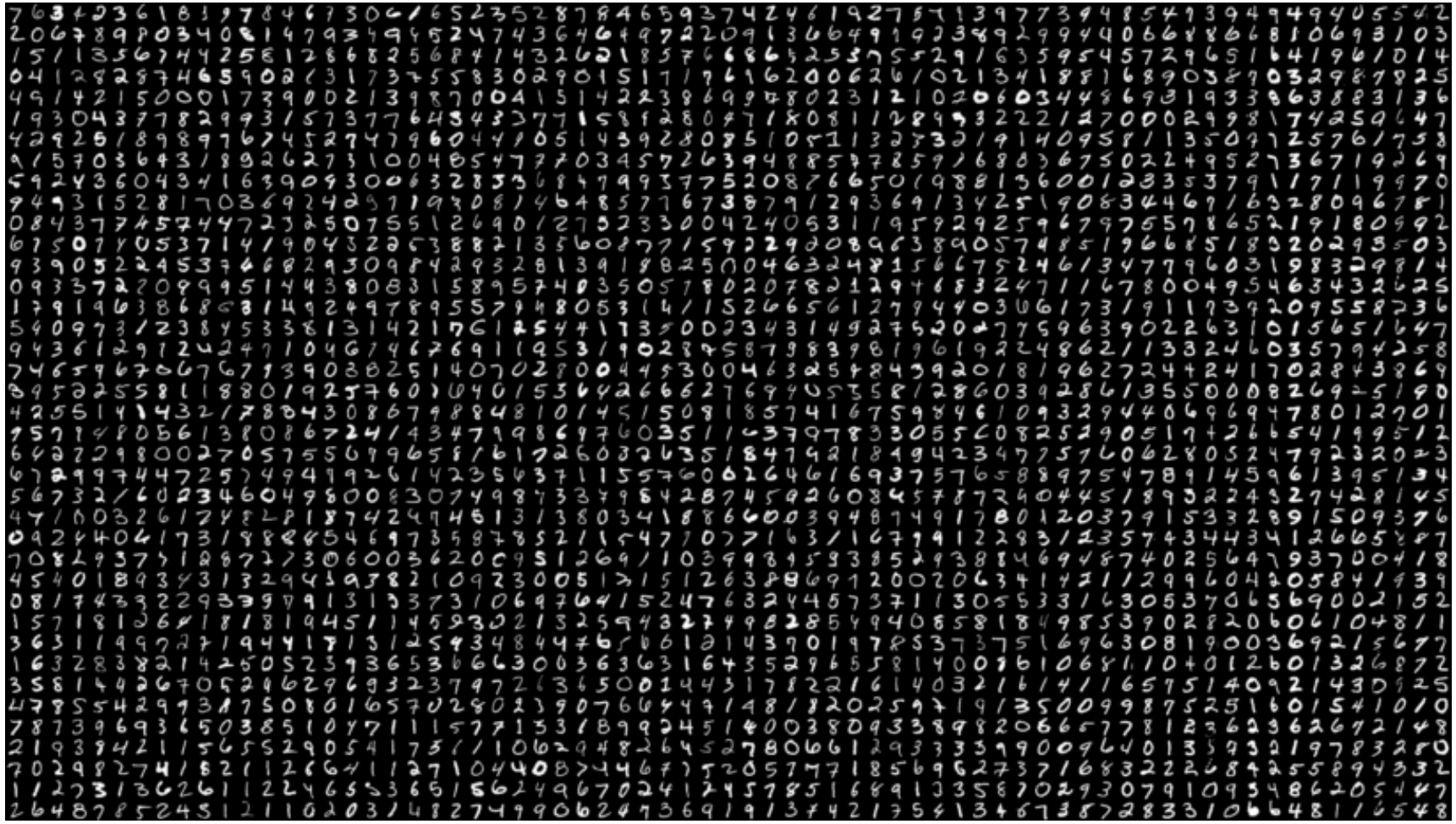
Hands-On `torch_tensor.ipynb`

Linear Regression

Hands-On



Loading Data, MNIST



Data Loading in PyTorch

Files, RAM, DB, …

class inherited from torch.utils.data.Dataset

- indexed (random) access to the samples e.g. (x, y)-pairs
- configure pre-processing steps (i.e. normalisation), etc.

see blog for custom datasets

torch.utils.data.DataLoader

- iterator over batches of samples: batch of x's, batch of y's
- shuffling of samples when iterator is reset
- parallelised access
- memory pinning
- etc.

see blog article for some explanation and additional recommendations

MNIST Dataset with PyTorch

Use datasets.MNIST to load 60’000 images for training and 10’000 for testing

```
import torch
from torchvision import datasets
from torchvision.transforms import ToTensor
import matplotlib.pyplot as plt
```

```
training_data = datasets.MNIST(
    root="data",
    train=True,
    download=True,
    transform=ToTensor()
)
```

for data downloaded to a subfolder of your working directory.

`transform`
for preprocessing pipeline
`ToTensor`
Transforms PIL images or numpy-arrays to tensor.

```
print(training_data)
```

```
Dataset MNIST
    Number of datapoints: 60000
    Root location: data
    Split: Train
    StandardTransform
Transform: ToTensor()
```

see torch.utils.data for further details on datasets and loading data

Loading MNIST with PyTorch

Elements are (x,y)-tuples with

x : a pytorch tensor with the 2d grayscale pixel values

y : an int with the label

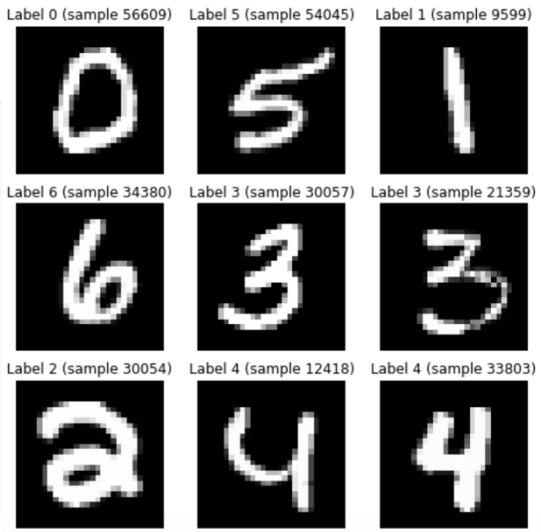
6th sample

```
print(len(training_data))
x, y = training_data[5]
print(type(x), x.shape, x.dtype)
print(type(y), y)
```

tensor with shape 1x28x28 with the first dimension corresponding to the color channel

```
60000
<class 'torch.Tensor'> torch.Size([1, 28, 28]) torch.float32
<class 'int'> 2
```

```
figure = plt.figure(figsize=(8, 8))
cols, rows = 3, 3
for i in range(1, cols * rows + 1):
    sample_idx = torch.randint(len(training_data), size=(1,)).item()
    img, label = training_data[sample_idx]
    figure.add_subplot(rows, cols, i)
    plt.title("Label %i (sample %i)"%(label,sample_idx))
    plt.axis("off")
    plt.imshow(img.squeeze(), cmap="gray")
plt.show()
```



DataLoader Class in PyTorch

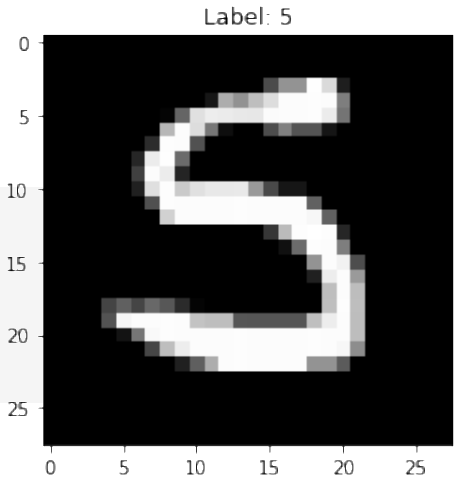
```
from torch.utils.data import DataLoader
train_dataloader = DataLoader(training_data, batch_size=64, shuffle=True)
```

```
data_train, labels_train = next(iter(train_dataloader))
print(f"Feature batch shape: {data_train.shape}")
print(f"Labels batch shape: {labels_train.shape}")
```

```
Feature batch shape: torch.Size([64, 1, 28, 28])
Labels batch shape: torch.Size([64])
```

tensor with shape (64x1x28x28) with the first dimension corresponding to the sample index in the batch

```
img = data_train[0].squeeze()
label = labels_train[0]
plt.imshow(img, cmap="gray")
plt.title(f"Label: {label}")
plt.show()
```



Hands-On

`load_mnist.ipynb`